BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 269 · Operating-system types

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which scheduler uses a fixed quantum?

Q2. What does starvation mean?

Q3. What matters most to a real-time task?

Q4. Rising-edge flip-flop Q=0. D pulses high between edges but is 0 at next rising edge. What is new Q?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

real-time — Deadline-bound response.

embedded — Dedicated device software.

distributed — Coordinated networked machines.

multi-user — Account-based sharing.

multitasking — Several processes progressing.

New learning

Understand — Real-time OS correctness depends on deadlines, not just average response speed.

Trace — Embedded and distributed refer to system architecture and purpose; each can overlap with real-time.

Evaluate — Multi-user concerns people/account isolation, whereas multitasking concerns processes.

Worked example

A safety-critical industrial controller has a fixed response deadline. Which OS type suits it?

1. Predict first — Predictable bounded response is required to meet safety deadlines.

2. Model and check — Explain why each value follows from the earlier step.

We Do · Practise with guidance

A university server supports many people with separate home directories. What OS characteristics?

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Multi-user for identity/isolation; multitasking for simultaneous services.
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Predict first

Many accounts require permissions and several running processes share CPU time.

Model and check

Explain why each value follows from the earlier step.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Match the five OS categories to an identifying property. [5]

Question 2

Explain OS requirements of a dedicated emergency-stop controller. [6]

Question 3

Explain why a networked cloud service can be distributed, multi-user and multitasking. [6]

Exit Ticket · Answer before checking

An embedded OS can never also be real-time.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Round Robin.

Q2. A task can be indefinitely denied execution.

Q3. Meeting the specified response deadline.

Q4. Q remains 0; only D at the active clock edge is sampled.

You Do pages · Independent questions

Q1. Real-time deadlines [1]; embedded dedicated [1]; distributed machines [1]; multi-user accounts [1]; multitasking processes [1].

Q2. Embedded purpose and restricted hardware [3]; real-time bounded response to critical deadline [3].

Q3. Distributed spans cooperating computers [2]; multi-user isolates independent accounts [2]; multitasking schedules multiple process workloads [2].

Exit Ticket · Model answer

False. OS type classifications overlap.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: